

Anchoring Effect, Loss Aversion, and Bidders' Behavior: New Evidence from NFT Auctions*

Lin William Cong[†]

Xiangchen Liu[‡]

Yu Wang[§]

December 3, 2024

Abstract

We study anchoring effects and loss aversion in the Non-Fungible Token market using comprehensive user-auction data from *CryptoPunks*. Using the last period's transaction price as the reference point, we separately measure anchoring effects and loss aversion for sellers and bidders. Our findings reveal that both sellers and bidders exhibit loss aversion, though the magnitude of loss aversion is significantly smaller for bidders. The analysis also confirms the presence of anchoring effects in both groups, with bidders displaying weaker effects than sellers, possibly due to signaling dynamics. Furthermore, we show that negotiations between sellers and bidders result in an intermediate level of loss aversion, falling between their individual measured levels. We also test whether experience influences loss aversion within both groups. For sellers, greater experience mitigates loss aversion, whereas for bidders, the results are mixed, with some evidence suggesting that experience may even increase their loss aversion in the NFT market.

*We are grateful to Levon Barseghyan, Xing Huang, Lawrence Jin, Siguang Li, Alejandro Martínez-Marquina, Kristoffer Nimark, Eswar Prasad, Seth Sanders, Adelson Teh, Zebang Xu, and Yiwei Zhang for their detailed comments and insightful suggestions. We also thank the participants of the Macro Lunch Seminar, Third Year Seminar of Cornell Economics Department, and the Asian Meeting of the Econometric Society in Singapore for their feedback.

[†]Lin William Cong: Cornell SC Johnson College of Business (Johnson) and NBER. Email: will.cong@cornell.edu.

[‡]Xiangchen Liu: Department of Consumer Science, University of Wisconsin-Madison. Email: xliu739@wisc.edu.

[§]Yu Wang: Department of Economics, Cornell University. Email: yw2322@cornell.edu.

1 Introduction

Loss aversion (Kahneman & Tversky, 1979) and the anchoring effect (Kahneman & Tversky, 1974) are notable sources of agents’ behavior biases in the field of social science. Loss aversion is the empirical observation that decisions often reflect a larger distaste for potential losses, compared to equal-sized gains, relative to a point of reference. the anchoring effect is the disproportionate influence on decision makers to make judgments that are biased toward an initially presented value. However, measuring these two behavioral biases is challenging, as different strategies and environmental settings can yield significantly different results (Gal & Rucker, 2018; Yechiam, 2019). For example, (Zeif & Yechiam, 2022) recently argues that loss aversion does not materialize for smaller losses.

In this paper, we aim to enhance the literature by simultaneously estimating anchoring effects and loss aversion in the Non-Fungible Token (NFT) market. This rapidly evolving digital ecosystem allows for the buying, selling, and trading of unique digital assets, such as art and collectibles, on blockchain platforms, thereby establishing ownership and provenance in the digital age (Wang, Li, Wang, & Chen, 2021). We utilize a novel user-action level dataset, *CryptoPunks*, to separately identify anchoring effects and loss aversion for bidders and sellers.

Documenting anchoring effects and loss aversion helps researchers, investors as well as policy makers study the NFT market, as a new emerging market. Early research in economics focused on introducing the NFT market and highlighting institutional facts (Bao & Roubaud, 2021; Nadini et al., 2021; Wang et al., 2021). Studies such as Oh, Rosen, and Zhang (2022) examine NFT market efficiency and information asymmetry, while Borri, Liu, and Tsyvinski (2022); Horky, Rachel, and Fidrmuc (2022); Kong and Lin (2021) investigated market prices and returns using empirical asset pricing approaches. Our paper, instead, investigate the NFT market through the lens of prospect theory, exploring whether investors in the NFT market exhibit similar behavioral biases as observed in traditional art markets. To the best of our knowledge, our study is the first to examine the NFT market from a behavioral economics perspective.

We consider the NFT market to be a more suitable environment for estimating anchoring effects and loss aversion for several reasons. First, the NFT market provides detailed data on individual sellers and bidders. In contrast to prior studies on anchoring effects (Beggs & Graddy, 2009; Graddy, Loewenstein, Mei, Moses, & Pownall, 2015), which typically rely on transaction prices that may not fully capture agents’ behavioral biases, the

NFT market offers high-frequency data that better reflects these dynamics. Similarly, for loss aversion, the NFT market provides richer data compared to slower-moving markets like real estate ([Genesove & Mayer, 2001](#)). Second, NFT auctions are algorithmically executed based on transparent bidding activity, without the intervention of brokers, auction experts, or auctioneers. This eliminates human biases and incentives that can distort price processes, yielding cleaner measurements of investor behavior (e.g., [Orley Ashenfelter and Kathryn Graddy \(2003\)](#) for expert bias, [Beggs and Graddy \(2009\)](#) for auctioneer bias, and [Rutherford, Springer, and Yavas \(2005\)](#) for housing brokers' incentives).

In this paper, enabled by a unique dataset at the user-auction level, we first estimate the loss aversion and anchoring effects of sellers, closely follow the methodology in [Bokhari and Geltner \(2011\)](#). We find that for sellers, when facing a potential loss, a 10 percent increase in the prospective loss leads to a 4.2 percent increase in the offering price. This yields an estimated loss aversion coefficient of 21, which is substantially higher than previous findings in other markets (e.g. 1.820-2.102 in a review paper by [Brown, Imai, Vieider, and Camerer \(2024\)](#)). Additionally, we observe an anchoring effect: after accounting for loss aversion, a 10% increase in sellers' offering prices over the market value results in an approximate 2.3% increase in final achieved selling prices. However, it is important to note that we could not distinguish whether the effects found here were due to signaling or anchoring, as sellers might have inside information and use higher prices to signal these features.

Conditional on the same reference point, we also for the first time estimate the loss aversion and anchoring for bidders. Our analysis reveals that bidders in the NFT market also exhibit behavioral biases, albeit to a lesser extent than sellers ([Chapman, Snowberg, Wang, & Camerer, 2024](#)). We find that when facing potential loss, a 10 percent increase in the difference between the previous sale price and the hedonic prediction results in a 1.5% increase in bidding prices, with a loss aversion coefficient of 7.5. This provides the first empirical evidence supporting the existence of the bidder's "endogenously generated reference point," as proposed by the literature on theoretical behavioral economics ([Kőszegi & Rabin, 2006](#); [Lange & Ratan, 2010](#); [Shunda, 2009](#)). Additionally, we observe an anchoring effect in bidders' behavior, where a 10% increase in bidding prices leads to a 0.6% increase in final transaction prices. This effect, while smaller than that observed for sellers, represents a purely psychological anchoring bias, if we assume bidders lack inside information about the assets.

We also use transaction prices as the outcome variable and find that loss aversion sig-

nificantly influences final transaction prices in the NFT market, reflecting a bargaining process between sellers and bidders. Our analysis reveals that when an NFT is facing a potential loss, a 10 percent increase in the positive difference between the previous sale price and the hedonic prediction leads to a 2.5 percent upward adjustment in the transaction price. The calculated loss aversion coefficient of 12.5 for final transaction prices falls between those of sellers (21) and bidders (7.5), suggesting that the ultimate price represents a middle ground between the two parties' behavioral biases.

We also compare the heterogeneity between groups with different experience within bidders and sellers. We measure experience using two proxies: the number of previous actions and time spent in the market by agents. For sellers, we find that experience significantly mitigates loss aversion. Specifically, the most experienced sellers exhibit a loss aversion coefficient of 18, substantially lower than the 41 observed for the least experienced. This trend persists in fresh actions, where we exclude the possibility that agents are influenced by previous actions or other agents' behavior. However, bidders show a different pattern: more experienced bidders demonstrate stronger reference dependence when using activity-based experience, particularly in loss situations. When using time spent in the market as a measure of experience, bidders show less clear patterns.

Our paper is positioned at the intersection of three main areas of literature: the study of illiquid asset markets, the Non-fungible Token (NFT) market, and prospective theory. Loss aversion and anchoring effects are well documented in the real estate housing market (Bokhari & Geltner, 2011; Bracke & Tenreyro, 2021; Genesove & Mayer, 2001; Graddy et al., 2015; Northcraft & Neale, 1987), art markets (Beggs & Graddy, 2009; Graddy et al., 2015), horse racing betting (Jullien & Salanie, 2000), experiment auctions (Ivanova-Stenzel & Seres, 2021), online auctions (Yasseri & Reher, 2022), cryptocurrency markets (Gurdgiev & O'Loughlin, 2020), credit lending market (Dougal, Engelberg, & Parsons, 2015) etc. However, prior studies rely solely on sales or listing data, and only separately examining anchoring and loss aversion effects on both buyers and sellers. . We contribute on this literature by making use of a novel NFT dataset with transaction-level details to overcome limitations of prior work relyingxxxxx. Our paper also contributes by providing empirical evidence on the existence of self-generated reference points for bidders, contributing to and supporting this literature (Kőszegi & Rabin, 2006; Lange & Ratan, 2010; Shunda, 2009).

Our paper also linked to the literature of potential moderators of reference dependence (Mrkva, Johnson, Gächter, & Herrmann, 2020). Empirical and theoretical literature on this topic has yielded mixed results. Some studies, such as Genesove and Mayer

(2001), J. A. List (2003), John A. List (2004), and J. List and Price (2013) support the notion that experience can moderate reference dependence. Conversely, Bokhari and Geltner (2011) found no evidence of such moderation, while Haigh and List (2005) and Pope and Schweitzer (2011) provide support for the opposite. In our research, we contribute to this literature by utilizing two measures, traders' activeness and the period between their first and current actions, as approximations of experience. By examining how experience interacts with loss aversion and the anchoring effect in a new financial market, we provide additional evidence and insights into the moderating role of experience.

The research on illiquid asset markets, including art markets (Lovo & Spaenjers, 2018; Mandel, 2009; Mei & Moses, 2002; Orley Ashenfelter & Kathryn Graddy, 2003), wine (Dimson, Rousseau, & Spaenjers, 2015), stamps (Dimson & Spaenjers, 2011), and sneakers, has received extensive attention. Our paper contributes to this literature by examining the price and returns of CryptoPunks, a representative NFT project, filling a significant gap in the literature. This complements the study by Kong and Lin (2021), which also focuses on CryptoPunks but with a different emphasis: empirical asset pricing versus our emphasis on agents' behavioral bias in the cryptocurrency market. Additionally, we utilize more detailed attribute data in our hedonic estimation.

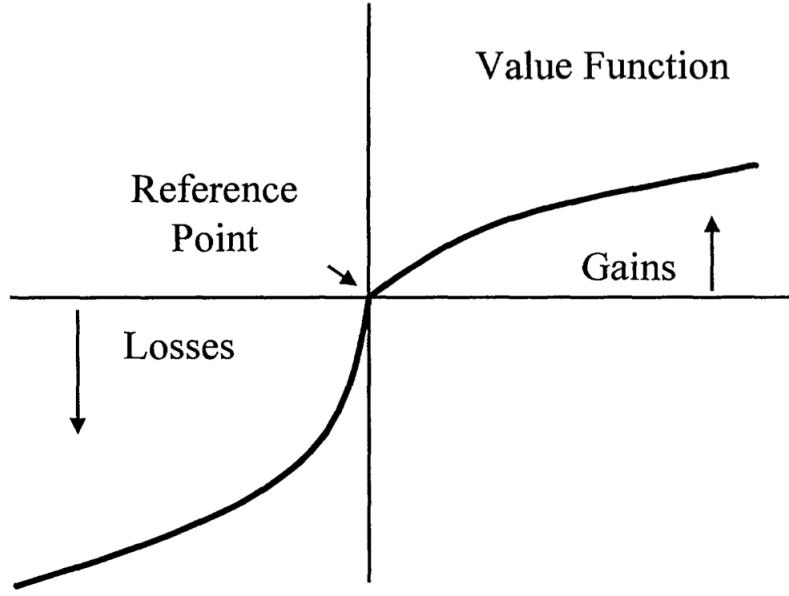
2 A Theoretical Framework of Reference Dependence and Loss Aversion

In the Prospect Theory proposed by Kahneman and Tversky (1974) and Kahneman, Knetsch, and Thaler (1990), there are three features that a utility function should be capturing with loss aversion. The first one is that both gain and loss are evaluated by a reference point. Second, with the equivalent size of gain and loss, a raise in utility with gain is less than a decline with loss. Third, changes in utility by gain or loss are decreasing with the sizes of gain and loss. We borrow a figure from Genesove and Mayer (2001) in Figure 1 to show these figures below.

To formalize the loss aversion in agent's utility function, we follow the theoretical framework in Bokhari and Geltner (2011), by assuming an increasing utility function $U(P)$ in the sale price P (namely, $U'(P) > 0$) and concave (i.e. $U''(P) < 0$) for all P but the reference price, R_p .¹ The agent maximizes her utility by setting an offering price in a potential

¹It is easy to see that the utility function is not differentiated when $P = R_p$.

Figure 1: An Example of Loss Aversion Utility Function



Note: This figure is borrowed from the Figure II in [Genesove and Mayer \(2001\)](#). The value function is referred to agent's utility function.

transaction

$$\max_P \mathbf{P}(P)U(P) + (1 - \mathbf{P}(P)) U_0, \quad (1)$$

where $\mathbf{P}(P)$ is the probability of goods successfully being sold and it is decreasing with the set price P (i.e. $\mathbf{P}'(P) < 0$). U_0 refers to the utility that the agent fails to sell her goods and we assume $U_0 < U(P)$. The first order condition with respect to the price generate

$$\mathbf{P}(P)U'(P) = \mathbf{P}'(P) (U_0 - U(P)), \quad (2)$$

which equates the potential benefit of raising sale price and the potential loss.

The loss aversion enters into the model through a parameter λ in agent's utility function $U(P)$. Specifically, her utility function is given by

$$U(P) = \begin{cases} (P - R_p) & \text{if } P \geq R_p \\ \lambda (P - R_p) & \text{if } P < R_p \end{cases}, \quad (3)$$

and $\lambda > 1$ is consistent with loss aversion. Even though this is a static single agent framework, we emphasize that we do not limit λ as a constant. Instead, λ , or λ_{it} , could vary

across different agents and time. This view is supported by the literature of moderators of loss aversion. Plugging the this functional form of utility function into the first order condition

$$\mathbf{P}(P) = -\mathbf{P}'(P) ((P - R_p) + U_0) \quad \text{if } P \geq R_p \quad (4)$$

$$\lambda \mathbf{P}(P) = -\mathbf{P}'(P) (\lambda (P - R_p) + U_0) \quad \text{if } P < R_p. \quad (5)$$

To see how loss aversion changes the pricing behavior of this agent, we further assume $\mathbf{P}''(P) = 0$. With this assumption, the comparative study can show that

$$\frac{\partial (P - R_p)}{\partial \lambda} > 0, \quad (6)$$

which indicates that the agent tends to raise sale price relative to the reference point when having loss aversion in her utility function. The more stronger the effect of loss aversion is (larger λ), the higher the offering price it will be relative to the reference price.

The reference point, R_p , can not be observed from the data directly. In fact, literature also discuss how agents update and set different reference dependence points (e.g. [Thakral and Tô \(2021\)](#) and [Guo \(2024\)](#) among others). With this being said, bidders might also anchor on the past transaction price as their reference point. Since bidders are not holding any goods in hand, nor they have paid any prices to purchase goods as well. Hence, intuitively λ_b for bidders should be zero, i.e. gain and loss will have the same impact on their behaviors and utility.

3 Empirical Strategies

3.1 Hedonic Regression: Predicting Prices by Attributes and Time

Hedonic models have widespread application in various financial markets, including housing markets (e.g. [Sopranzetti \(2010\)](#)), collectible markets (e.g. [Wyburn and Roach \(2012\)](#)), and even the computation of Consumer Price Index (CPI) indices (e.g. [Fixler, Fortuna, Greenlees, and Lane \(1999\)](#)). By considering the characteristics and transaction details of items, hedonic regression can effectively predict prices for each item in different time periods.

In this study, we employ hedonic regression to predict the prices of CryptoPunks by regressing the logarithm of the transaction prices (π_{it}) on the types and attributes of the

CryptoPunks ($\mathbf{X}_i$), while controlling for fixed effects at the monthly level (δ_t)

$$\pi_{it} = \mathbf{X}_i\beta + \delta_t + \varepsilon_{it}, \quad (7)$$

where subscript i refers each CryptoPunk and t denotes month. To capture the potential value associated with individual attributes or characteristics of each CryptoPunk, we include all 92 different attributes as indicator variables in the regression. The coefficients (β) associated with these variables quantify the potential impact of each attribute on the predicted price.

3.2 Jointly Testing Anchoring Effect and Loss Aversion

Loss Aversion. We closely follow the literature (Genesove and Mayer (2001) and Bokhari and Geltner (2011)) to estimate the potential loss aversion of sellers and bidders in the CryptoPunks market, under the assumption that both of their reference point is set to be the previous transaction price. Specifically, we employ the following regression

$$P_{it} = \mu\pi_{it} + \lambda(P_{is} - \pi_{it}) \cdot \mathbf{1}_{\{P_{is} - \pi_{it} < 0\}} + \xi(P_{is} - \pi_{is}) + \varepsilon_{it} \quad (8)$$

where the left hand side variable, P_{it} , represents either the sale price, offering price, or bidding price of Punk i at period t in logarithm. Meanwhile, P_{is} denotes the price of item i at its previous transaction, serving as the reference point. The concept of reference dependence is captured by the term $(P_{is} - \pi_{it})$, and the parameter λ quantifies the extent to which the difference between the previous period's price and the hedonic predicted price influences the dependent variable. The ratio of the estimated λ when $(P_{is} - \pi_{it}) < 0$ to $(P_{is} - \pi_{it}) > 0$ indicates the magnitude of loss aversion. Additionally, we incorporate the residual $(P_{is} - \pi_{is})$ from the last sold price to account for unobserved quality changes and any over- or under-payment.

By applying Equation (8) to sellers' offering prices, we divide the full sample into a gain sample (where $(P_{is} - \pi_{it}) > 0$) and a loss sample (where $(P_{is} - \pi_{it}) < 0$). If the estimated λ^- in the loss sample is larger than λ^+ in the gain sample, we conclude that sellers experience loss aversion, with its magnitude given by $\frac{\lambda^-}{\lambda^+}$. Since the loss aversion literature has found widespread evidence of this behavior across various markets, our prior assumption is that $\frac{\lambda^-}{\lambda^+} > 1$.

Applying the same empirical strategy to a "gain sample" and a "loss sample" of bid-

ders' actions, we do not expect a significant difference between λ^- and λ^+ , as bidders intuitively should not experience loss aversion since they have not yet paid to purchase the Punks. In fact, [Beggs and Graddy \(2009\)](#) applied Equation (8) to transaction price data from art auctions, combining both gain and loss samples, and interpreted the significant λ as evidence of bidders' anchoring effects.

Empirically, there is also no evidence supporting the existence of loss aversion on the bidder's side. However, previous theoretical research (see [Kőszegi and Rabin \(2006\)](#), [Shunda \(2009\)](#), and [Lange and Ratan \(2010\)](#)) has proposed the potential for the endogenous formation of bidders' reference points. In particular, [Lange and Ratan \(2010\)](#) predicts that loss-averse bidders tend to overbid, as a winning probability lower than the reference point inflicts significantly greater harm. This theoretical foundation for the existence of bidders' loss aversion may help explain unexpected empirical results.

Finally, we also apply the same empirical strategy to the realized transaction prices to investigate how the bargaining process between bidders and sellers affects the estimated loss aversion. We hypothesize that a smaller estimated ratio of $\frac{\lambda^-}{\lambda^+}$, reflecting the absence of loss aversion on the bidder's side, could potentially reduce part of the observed loss aversion on the seller's side.

Anchoring Effects. We closely follow [Bokhari and Geltner \(2011\)](#) to identify anchoring effects in the CryptoPunks market. Specifically, we first measure, $\hat{\varepsilon}_{it}$, the residual from Equation (8), then regress it on the transaction prices in the Hedonic regression

$$\pi_{it} = \mathbf{X}_i\beta + \gamma\hat{\varepsilon}_{it} + \delta_t + u_{it}, \quad (9)$$

where u_{it} is the error term. If γ is statistically significantly different from 0, it indicates that the components of sellers' offering or bidders' bidding prices, which cannot be explained by loss aversion, have explanatory power over the transaction prices.

One potential concern with testing anchoring effects using Equation (9) is that it may capture signaling effects, where sellers intentionally set a price that deviates from their typical offering. To disentangle these confounding signaling effects, we estimate Equation (9) using residuals from both sellers and bidders. If private information about Crypto Punks is only observed by sellers, the estimated $\hat{\gamma}$ will differ when using bidders' residuals versus sellers' residuals.

4 Data and Market Structure

4.1 Crypto Punk Data

This paper utilizes a distinctive dataset at the user-auction level, obtained from CryptoPunks, to examine behavioral biases exhibited by agents. CryptoPunks represents one of the earliest instances of "Non-Fungible Tokens" (NFTs) on the Ethereum blockchain and served as the inspiration for the ERC-721 standard, which underpins the majority of digital art and collectibles. The CryptoPunks collection comprises 10,000 individually generated characters commonly referred to as "punks." Each punk is a 24x24 pixel art image that is algorithmically generated. These punks are unique, with no two being identical, and each one can be officially owned by a single individual on the Ethereum blockchain. ².

Our dataset primarily originates from Larva Labs, the entity behind the creation of the CryptoPunks project. We have compiled comprehensive information regarding all CryptoPunks, encompassing attributes, types ³ for hedonic regression, historical auction records for testing the anchoring effect and loss aversion, ownership details, and creatorship information. The historical auction records within our dataset include seven distinct types of investors' actions: "Sold," "Bid," "Bid Withdrawn," "Offered," "Offer Withdrawn," "Transfer," and "Claimed." The majority of these actions are self-explanatory, although two actions may require clarification: "Transfer" and "Claimed." The "Transfer" action indicates that the ownership of a particular punk has been transferred from one account to another, without any observable transactions between the parties involved. This transfer may occur without any monetary exchange, suggesting a transfer of ownership free of charge. On the other hand, the "Claimed" action denotes the initial creation of a punk. When a punk is claimed, it means that it was generated and entered into circulation for the first time.

Our sample encompasses the period from June 23rd, 2017, which marks the inception of the CryptoPunk market, to July 21st, 2021. While the duration of our sample may appear

²Initially, all punks were freely claimable by anyone with an Ethereum wallet. However, they are now traded within the CryptoPunks marketplace, which is embedded within the Ethereum blockchain. This marketplace enables users to purchase, place bids on, and offer punks for sale. In general, data of Non-fungible tokens is of free access if one is able to connect to blockchain. In addition, there are also a few online platforms providing auction information in the NFT market, such as SuperRare, OpenSea and NonFungible. Some of them also serve as trading platforms on which investors can trade there (e.g., SuperRare, openSea). Non-Fungible does not allow people to trade but provides transaction history and generates analysis of the NFT market.

³On Larva Labs, the punks are categorized into five distinct groups: Alien, Ape, Zombie, Female, and Male. Furthermore, there exist 87 different attribute types that individual punks may possess in their respective images, such as Choker, Pilot Helmet, or Tiara.

Table 1: Action Summary Statistics

Action Types	Number	Percent	Cum. Percent
Bid	22525	18.12	18.12
Bid Withdrawn	13101	10.54	28.66
Claimed	10000	8.05	36.71
Offer Withdrawn	6856	5.52	42.22
Offered	45910	36.94	79.16
Sold	14650	11.79	90.95
Transfer	11253	9.05	100.00
Total	124295	100.00	

relatively short in comparison to the extensive historical records available for traditional art auctions, it is important to highlight the unparalleled level of detail captured in our dataset. We have compiled comprehensive auction bidding and transaction histories for all NFT assets within the CryptoPunk market, surpassing the level of detail observed in previous empirical studies within the anchoring effect/loss aversion literature. This dataset provides a unique opportunity to study the CryptoPunks market comprehensively and draw insights from the entire range of auction activities. To provide a snapshot of the historical auction records during our sample period, we present summary statistics in Table 1.

As depicted in Table 1, the entire sample comprises 124,295 actions conducted by bidders and sellers within the CryptoPunks market. Among these actions, the category "Offered" represents the largest proportion, accounting for nearly 37% of the entire sample. Additionally, the category "Bid" constitutes slightly over 18% of the total actions. When comparing the behavior of bidders and sellers, it is evident that bidders tend to withdraw their bids more frequently than sellers. Furthermore, the sample includes a notable number of "Transfer" actions, which aligns with the reported trend indicating the growing popularity of Non-Fungible Tokens as gifts among blockchain and cryptocurrency enthusiasts.

In Tables 2, 3, and 4, we delve deeper into the three key actions of interest in auctions: "Sold," "Bid," and "Offered". Variables ends with "DP" indicates that the prices are measured in dollars, while with "EP" represents prices in Ethereum. Additionally, we divide our complete sample into gain and loss subsamples. To determine the gains and losses, we evaluate the difference between the hedonic predicted price and the transaction price in the previous period. If the hedonic predicted price is lower than the transaction price in the last period, the observation falls into the loss subsample. This implies that investors

Table 2: Summary Statistics: "Sold"

<i>Panel A: Full Sample</i>	Num.	Mean	SD	Min	Max
sold(USD)	14648	27666.50	104371.24	0	7580000
sold(Ether)	14627	14.41	55.47	0	4200
<i>Panel B: Gain Sample</i>	Num.	Mean	SD	Min	Max
sold(USD)	9336	64770.31	416321.85	0	7579998
sold(Ether)	9226	33.05	230.44	0	4200
<i>Panel C: Loss Sample</i>	Num.	Mean	SD	Min	Max
sold(USD)	12924	35574.74	69971.78	0	1760000
sold(Ether)	13007	19.53	35.66	0	667

Table 3: Summary Statistics: "Bid"

<i>Panel A: Full Sample</i>	Num.	Mean	SD	Min	Max
bid(USD)	22525	33146.73	127583.51	0	7580000
bid(Ether)	21759	17.56	68.28	0	4200
<i>Panel B: Gain Sample</i>	Num.	Mean	SD	Min	Max
bid(USD)	5102	45484.26	159470.58	0	7579998
bid(Ether)	4912	22.84	89.42	0	4200
<i>Panel C: Loss Sample</i>	Num.	Mean	SD	Min	Max
bid(USD)	9494	21229.02	49761.41	0	1680001
bid(Ether)	9421	11.78	24.24	0	667

potentially face a loss if they consider the last period's selling price as a reference point. In our sample, we exclude the observations of "Bid" and "Offered" if there are no subsequent transactions recorded after these actions. In other words, if there are no additional "Sold" actions following the "Bid" or "Offered" actions, we remove those specific observations from the analysis.

4.2 Crypto Punk Market Structure

The Non-Fungible Token (NFT) market presents a distinct advantage when examining the anchoring effect and loss aversion, particularly when compared to traditional art and real

Table 4: Summary Statistics: "Offered"

<i>Panel A: Full Sample</i>	Num.	Mean	SD	Min	Max
offer(USD)	45867	134546.49	3497289.55	0	537340032
offer(Ether)	45848	166.07	9303.11	0	1000000
<i>Panel B: Gain Sample</i>	Num.	Mean	SD	Min	Max
offer(USD)	18232	79291.05	1235558.89	0	134009960
offer(Ether)	17738	48.67	1201.49	0	88890
<i>Panel C: Loss Sample</i>	Num.	Mean	SD	Min	Max
offer(USD)	18531	43930.77	177102.90	0	14559988
offer(Ether)	19004	106.39	8196.79	0	1000000

estate markets. One key advantage is its concise market structure. Unlike the traditional markets, the NFT market operates on a blockchain, eliminating the need for intermediaries such as experts and auctioneers. This streamlined structure mitigates concerns related to measurement errors caused by disregarding brokers and bidders in the real estate market, as well as experts and auctioneers in the traditional art market.

As highlighted in a survey paper by [Orley Ashenfelter and Kathryn Graddy \(2003\)](#), the art auction market involves the participation of experts and auctioneers who can potentially influence auction outcomes. In this market, experts, who are typically appointed by auction houses, provide pre-sale estimates for the collectibles that will be auctioned. These estimates are made public and are accessible to all bidders and the seller. Although experts generally strive to provide unbiased estimates, research conducted by [Beggs and Graddy \(1997\)](#) revealed systematic under- and over-predictions in their assessments. Furthermore, [Orley Ashenfelter and Kathryn Graddy \(2003\)](#) argued that auctioneers have incentives to lower the estimated price in order to attract a larger pool of bidders, as supported by certain institutional factors.

Trading in the NFT market offers a simpler process compared to traditional auction houses. OpenSea, one of the most renowned NFT trading platforms, provides comprehensive guidelines on how to sell NFTs. Sellers have the flexibility to choose between a fixed-price sale or setting up a timed auction for their items. Additionally, they can opt for either an English auction or a Dutch auction format. In an English auction, sellers establish a starting price, a reserve price, and the duration of the auction. At the auction's conclusion, the highest bidder wins the item if their bid surpasses the seller's reserve price. On the other hand, in a Dutch auction, the seller initiates the sale at a higher price, which then decreases progressively until a bidder decides to purchase the item. Sellers also determine the starting price and duration for these auctions. It is worth noting that fixed-price sales and Dutch auctions document the sellers' offering price, while English auctions record the bids made by bidders.

5 Empirical Results: Measured Anchoring Effect and Loss Aversion

5.1 Hedonic Regression

In this section, we illustrate the results from the hedonic regression in Equation (7). We observe that the price variations can be effectively explained by the combination of characteristics and time dummies in the hedonic regression, yielding a R^2 value of 0.941. Additionally, while some of the observable characteristics demonstrate high statistical significance, it is important to note that not all of them exhibit such significance. Our interpretation favors the idea that only popular and noticeable characteristics, such as *blue eye shadow*, *choker*, and *nerd glasses*⁴, have a significant positive impact on the prices of NFTs. The market price index is illustrated in Figure 2.

Figure 2: Hedonic Price Index in the NFT Market

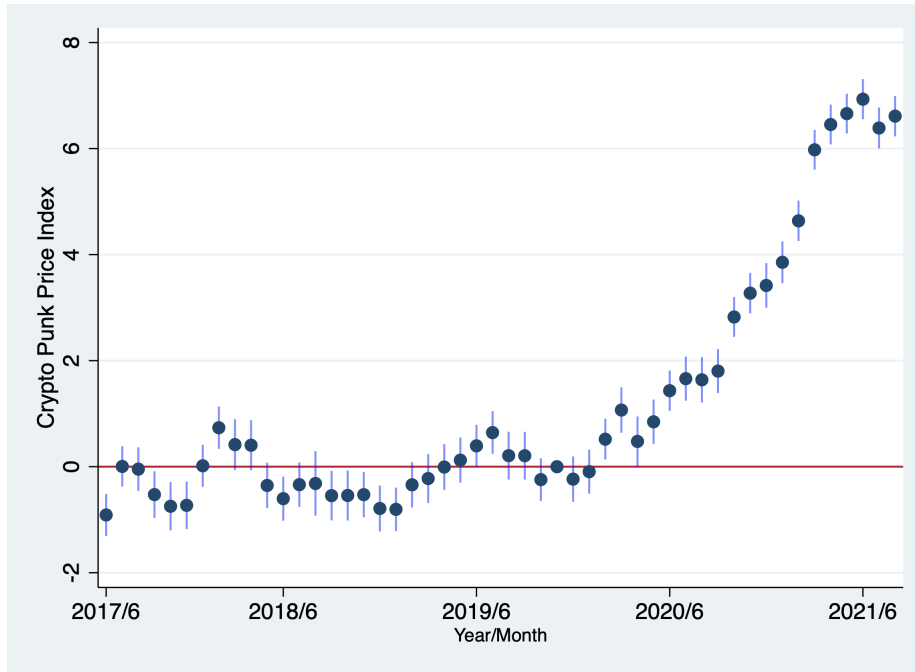


Figure 2 illustrates the market dynamics during our sample period. From June 2017 to June 2019, the market price of NFTs was relatively low, characterized by significant fluctuations. During this phase, the NFT market had yet to capture significant attention. However, as news media reports about the NFT market surged towards the end of 2019, the market witnessed a notable increase in prices. This upward trend was fueled by a growing number of investors entering the market, leading to an expanded demand for NFT assets.

⁴The increasing media coverage on this topic highlights how "nerd" is becoming a new trend.

The onset of the Covid-19 pandemic further amplified this trend. With people increasingly turning to online interactions, it comes as no surprise that electronic arts gained significant popularity among online communities.

5.2 Sellers' Loss Aversion

Table 5 presents the estimation results of sellers' loss aversion, in which Panel A, B and C are corresponding to testing sellers' loss aversion using full sample, gain sample and loss sample, respectively. To capture the potential effect of recent events on decision-making (?), we gradually narrow each subsample in each regression so that the offering actions occur closer to the last transaction. We calculate the 25th, 50th, and 75th percentiles of the time interval between the week of the offering actions and the week of the last transaction in the full sample. The results show that 25% of offering actions are made within one week, 50% within three weeks, and 75% within fifteen weeks.⁵

We initially regress offering prices against the last transaction prices using the full sample. The estimated coefficient of 0.05, which is statistically significant, suggests that sellers exhibit reference dependence, using the last transaction price as a benchmark. Specifically, a 10 percent increase in the positive difference between the previous sale price and the predicted hedonic price leads to a 0.5% increase in the offering price relative to the previous sale price. Furthermore, this reference dependence among sellers intensifies as the holding period shortens.

Additionally, our analysis reveals that the estimated coefficient in the loss sample (Panel C) is greater than that in the gain sample (Panel B). This indicates that sellers are more influenced by losses than by gains. Specifically, in the loss sample, a 10 percent increase in a prospective loss leads to a 4.2 percent increase in the offering price, whereas in the gain sample, the corresponding increase is only 0.2 percent. This disparity underscores the psychological phenomenon of loss aversion, where sellers tend to set higher prices when facing a potential loss compared to when they anticipate a gain.

We further quantify this by dividing the loss coefficient by the gain coefficient, resulting in a loss aversion coefficient of 21. This estimated value is notably higher than those reported in previous literature (e.g. from 1.8 to 2.1 in [Brown et al. \(2024\)](#)). The larger loss aversion in this context may be attributed to the heightened price volatility in the NFT mar-

⁵Literature of collectible markets tends to use the year as the control variable. However, transactions happen much more frequently in the NFT market when compared to traditional collectible markets (but still far less frequent than the stock market). Hence, we think it could be better to use weeks as the control variable.

Table 5: Sellers' Loss Aversion in Dollars

	(1) All	(2) < 16 weeks	(3) < 4 weeks	(4) < 2 weeks
<i>Panel A: Full Sample</i>				
Reference Dependence	0.05*** (0.00)	0.09*** (0.01)	0.42*** (0.03)	0.59*** (0.05)
Hedonic Price	0.93*** (0.00)	0.93*** (0.00)	0.92*** (0.00)	0.90*** (0.00)
Residuals from Last Sale	0.09*** (0.01)	0.02 (0.01)	-0.25*** (0.03)	-0.33*** (0.05)
Holding Period	0.00*** (0.00)	0.00 (0.00)	-0.01 (0.01)	-0.03 (0.02)
Observations	25923	19675	13703	9976
<i>Panel B: Gain Sample</i>				
Reference Dependence	0.02*** (0.01)	0.07*** (0.02)	0.41*** (0.04)	0.49*** (0.06)
Hedonic Price	0.94*** (0.00)	0.95*** (0.00)	0.92*** (0.00)	0.91*** (0.00)
Residuals from Last Sale	0.02*** (0.01)	-0.04** (0.02)	-0.34*** (0.04)	-0.37*** (0.06)
Holding Period	0.00*** (0.00)	0.02*** (0.00)	0.07*** (0.01)	0.09*** (0.02)
Observations	18232	12566	7999	5415
<i>Panel C: Loss Sample</i>				
Reference Dependence	0.42*** (0.04)	0.61*** (0.04)	1.04*** (0.07)	1.19*** (0.11)
Hedonic Price	0.92*** (0.00)	0.92*** (0.00)	0.92*** (0.00)	0.91*** (0.00)
Residuals from Last Sale	0.68*** (0.03)	0.46*** (0.04)	0.15** (0.06)	0.03 (0.10)
Holding Period	-0.00 (0.00)	-0.04*** (0.00)	-0.10*** (0.01)	-0.14*** (0.02)
Observations	7691	7109	5704	4561

Note: The dependent variable is sellers' offering prices. All measures are in dollar term. panel A shows the OLS regression results using all sellers' offering prices. Panel B shows the OLS regression results using sellers' offering prices when predicted prices are estimated higher than previous sale price at the time of offering. Panel C shows the OLS regression results using sellers' offering prices when predicted prices are estimated lower than previous sale price at the time of offering. Different samples are constructed by distribution of the holding period. The first to third quartile of holding period is 2, 4 and 15 weeks. All estimates use dollar term. Standard errors in parentheses. * : ($p < 0.10$), ** : ($p < 0.05$), *** : ($p < 0.01$).

ket, which could amplify the perceived risk and consequently the degree of loss aversion among bidders.

5.3 Bidders' Loss Aversion

We apply the same empirical strategy used to identify sellers' loss aversion to examine whether bidders may also experience loss aversion. Table 6 presents the results in a similar format: Panel A implements the exercise with the full sample, Panel B focuses on the gain sample, and Panel C examines the loss sample. Similarly, we also calculate the 25th, 50th, and 75th percentiles of the time interval between the week of the bidding actions and the week of the last transaction in the full sample. The results indicate that 25% of bidding actions occur within two weeks, 50% within nine weeks, and 75% within forty-six weeks.

Panel A shows that bidders are statistically significantly influenced by the last transaction prices, after adjusting for hedonic regression. Specifically, a 10 percent increase in the positive difference between the previous sale price and the hedonic prediction leads to a 0.3% upward adjustment in bidding prices towards the previous sale price. Moreover, this reference dependence among bidders becomes more pronounced as the holding period decreases.

We further explore the impact of "loss" versus "gain" on bidding prices. In the loss sample, a 10 percent increase in the difference between the previous sale price and the hedonic prediction results in a 1.5% increase in bidding prices, compared to a 0.2% increase in the gain sample. Although the coefficient is not statistically significant across all time frames (as seen in columns (1) and (2) for within 46 weeks), the results are statistically significant within the 9 and 2-week periods (columns (3) and (4)). The observed trend and effect remain consistent, suggesting that bidders, who do not own the items, still exhibit some form of loss aversion.

One potential explanation for the existence of bidders' loss aversion is that they may internalize sellers' loss aversion. Additionally, bidders themselves experience loss aversion, with their reference point being a lottery that reflects uncertain payoff outcomes. Their gains and losses must be evaluated against this lottery as a reference point. Thus, if bidders believe that sellers are only willing to sell the Crypto Punks when the transaction price exceeds the typical price, they will likely increase their bids more in the loss sample compared to the gain sample, using the last transaction price as their reference point.

When comparing the loss aversion coefficient between bidders and sellers, we find

Table 6: Bidders' Loss Aversion in Dollars

	(1) All	(2) < 47 weeks	(3) < 10 weeks	(4) < 3 weeks
<i>Panel A: Full Sample</i>				
Reference Dependence	0.03* (0.02)	0.08*** (0.03)	0.19*** (0.07)	0.37** (0.16)
Hedonic Price	1.02*** (0.01)	1.02*** (0.01)	1.04*** (0.01)	1.03*** (0.02)
Residuals from Last Sale	0.13*** (0.03)	0.09*** (0.03)	-0.06 (0.07)	-0.26 (0.16)
Holding Period	-0.00*** (0.00)	0.00* (0.00)	0.01 (0.01)	-0.01 (0.03)
Observations	6577	4944	3374	1868
<i>Panel B: Gain Sample</i>				
Reference Dependence	0.02 (0.02)	0.08*** (0.03)	0.27*** (0.09)	0.34** (0.15)
Hedonic Price	1.03*** (0.01)	1.03*** (0.01)	1.04*** (0.01)	1.03*** (0.02)
Residuals from Last Sale	0.11*** (0.03)	0.06* (0.03)	-0.14* (0.08)	-0.24 (0.15)
Holding Period	-0.01*** (0.00)	0.00 (0.00)	0.04*** (0.01)	0.11** (0.05)
Observations	5102	3579	2285	1170
<i>Panel C: Loss Sample</i>				
Reference Dependence	0.15 (0.14)	0.23 (0.16)	0.53*** (0.20)	0.92* (0.50)
Hedonic Price	1.03*** (0.02)	1.04*** (0.02)	1.06*** (0.03)	1.04*** (0.04)
Residuals from Last Sale	0.50*** (0.10)	0.43*** (0.12)	0.12 (0.18)	-0.32 (0.40)
Holding Period	0.00* (0.00)	0.01 (0.01)	-0.02 (0.02)	-0.02 (0.06)
Observations	1475	1365	1089	698

Note: The dependent variable is bidders' bidding prices. All measures are in dollar term. Panel A shows the OLS regression results using all bidders' bidding prices. Panel B shows the OLS regression results using bidders' bidding prices when predicted prices are estimated lower than previous sale price at the time of bidding. Panel C shows the OLS regression results using bidders' bidding prices when predicted prices are estimated lower than previous sale price at the time of bidding. Different samples are constructed by quartile distribution of the holding period. The first to third quartile of holding period is 2, 9 and 46 weeks. All estimates use dollar term. Standard errors in parentheses. *: ($p < 0.10$), **: ($p < 0.05$), ***: ($p < 0.01$).

that the loss aversion coefficient for bidders is 7.5, which is smaller than the coefficient of 21 observed for sellers. This difference may be attributed to the lesser impact of behavioral biases in self-generated reference dependence among bidders, as opposed to the more substantial effects seen with true ownership among sellers.

5.4 The Impact of Loss Aversion on Final Transaction Prices

In the previous analyses, we demonstrated that both sellers and bidders exhibit loss aversion in the NFT market. However, it remains unclear how the bargaining process between these two parties influences the final transaction price. In this section, we test the effect of loss aversion on the final transaction price, using the final achieved price as the dependent variable. Table 7 presents the results, which are structured similarly: the sample is divided based on holding periods—Punks sold within 21 weeks (Column 2), five weeks (Column 3), and one week (Column 4)—representing the third, second, and first quartiles of holding periods, respectively.

The patterns observed here generally mirror the results found for sellers and bidders. In the full sample analysis, we find that a 10 percent increase in the positive difference between the previous sale price and the hedonic prediction results in a 0.4 percent upward adjustment in the transaction price toward the previous sale price. This effect becomes more pronounced as the holding period decreases. In the loss sample, reference dependence yields a coefficient of 0.25. These results provide further evidence that loss aversion influences the final transaction price. However, since bidders exhibit less pronounced loss aversion compared to sellers, the final transaction prices appear to be a reflection of the bargaining process between the two parties, balancing the behavioral biases of both sides. When comparing the coefficients for the gain sample, which stands at 0.2 percent, we calculate a loss aversion coefficient of 12.5 for the final transaction price. This figure is larger than that for bidders but smaller than that for sellers, providing further evidence that the final transaction price reflects a middle ground between the loss aversion of sellers and bidders.

We repeat all of the empirical exercises described above using prices in Ether as the unit of measurement. The main results remain consistent with our previous findings, and the detailed outcomes can be found in the Appendix, along with the results of the placebo test.

Table 7: Price Anchoring Effect in Dollars

	(1) All	(2) < 22 weeks	(3) < 6 weeks	(4) < 2 weeks
<i>Panel A: Full Sample</i>				
Reference Dependence	0.04*** (0.01)	0.09*** (0.02)	0.23*** (0.04)	0.42*** (0.09)
Hedonic Price	1.00*** (0.00)	1.00*** (0.00)	1.00*** (0.01)	0.99*** (0.01)
Residuals from Last Sale	0.07*** (0.01)	0.02 (0.02)	-0.11** (0.04)	-0.24*** (0.09)
Holding Period	0.00*** (0.00)	0.00 (0.00)	0.01 (0.01)	-0.03 (0.03)
Observations	8777	6668	4583	2986
<i>Panel B: Gain Sample</i>				
Reference Dependence	0.02** (0.01)	0.07*** (0.02)	0.27*** (0.04)	0.34*** (0.09)
Hedonic Price	1.01*** (0.00)	1.01*** (0.00)	1.00*** (0.01)	1.00*** (0.01)
Residuals from Last Sale	0.02** (0.01)	-0.03* (0.02)	-0.23*** (0.04)	-0.27*** (0.09)
Holding Period	0.00*** (0.00)	0.01*** (0.00)	0.06*** (0.01)	0.04 (0.03)
Observations	6807	4897	3118	1877
<i>Panel C: Loss Sample</i>				
Reference Dependence	0.25*** (0.08)	0.33*** (0.10)	0.46*** (0.15)	0.96*** (0.25)
Hedonic Price	0.98*** (0.01)	0.98*** (0.01)	0.99*** (0.01)	0.98*** (0.01)
Residuals from Last Sale	0.49*** (0.06)	0.40*** (0.08)	0.29** (0.12)	-0.06 (0.23)
Holding Period	-0.00** (0.00)	-0.01** (0.00)	-0.07*** (0.02)	-0.05 (0.05)
Observations	1970	1771	1465	1109

Note: The dependent variable is the items' sale prices. All measures are in dollar term. Panel A shows the OLS regression results using sale prices. Panel B shows the OLS regression results using sale prices when predicted prices are estimated higher than previous sale price. Panel C shows the OLS regression results using sale prices when predicted prices are estimated lower than previous sale price. Different samples are constructed by quartile distribution of the holding period. The first to third quartile of holding period is 1, 5 and 21 weeks. All estimates use dollar term. Standard errors in parentheses. * : ($p < 0.10$), ** : ($p < 0.05$), *** : ($p < 0.01$).

5.5 The Impact of Anchoring on Transaction Prices

In this section, we employ the methods of [Bokhari and Geltner \(2011\)](#) to test whether sellers' offering prices and bidders' bidding prices influence the final achieved transaction prices after accounting for loss aversion. We use the residuals from Equation (8) as a right-hand side variable in the hedonic regression equation (7), where the coefficient of the residual captures how final transaction prices anchor and adjust based on sellers' offering prices and bidders' bidding prices.

Table 8: Effects of Sellers' Offering Prices on Transaction Prices

	(1) log(prices)
Anchoring/Signaling	0.23*** (0.01)
<i>N</i>	25882
Standard errors in parentheses	
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$	

Table 8 shows the hedonic regression of transaction prices, including residuals from sellers' loss aversion regression. We find that the anchoring effect is statistically significant, indicating that a 10% increase in sellers' offering prices over the market value, after accounting for loss aversion, results in an approximate 2.3% increase in final achieved selling prices. These results align with those of [Bokhari and Geltner \(2011\)](#), though they reported larger anchoring and adjustment effects using real estate data. The smaller coefficient found in our study might be attributed to the more complete and transparent information provided by the NFT market, where agents can correct their behavioral biases through learning.

Table 9: Effects of Bidders' Offering Prices on Transaction Prices

	(1) log(prices)
res_bidder_hold	0.06*** (0.01)
<i>N</i>	6544
Standard errors in parentheses	
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$	

It is important to note that [Bokhari and Geltner \(2011\)](#) could not distinguish whether the effects found in their analysis were due to signaling or anchoring, as sellers might have inside information and use higher prices to signal these features. However, since we use NFT data, which includes information from both sellers and bidders, we can test whether

final transaction prices also anchor on bidders’ bidding prices. As bidders do not possess inside information about the assets, and assume their bidding prices are not influenced by sellers’ offering prices if sellers aim to signal specific features of the asset, the effect of bidding prices over the market value and loss aversion on final transaction prices reflects purely psychological anchoring and adjustment effects. Table 9 presents the results from this analysis. Although the magnitude of anchoring on bidding prices is smaller than that on sellers’ offering prices, we still find statistically significant results. This implies that a 10% increase in bidding prices, after accounting for the market average and loss aversion, results in about a 0.6% increase in final achieved prices. Even if the influence of it is possible that sellers’ offering prices on final prices can largely be attributed to signaling, the effects observed for bidders clearly reflect anchoring effects. Although the anchoring effect is smaller than the loss aversion effect discussed previously, it suggests that not only does loss aversion exist in this auction market, but both loss aversion and anchoring biases are present throughout the bargaining process.

6 Experience as the Moderator: Is Experience Helpful?

In this section, we examine whether experience in offering and bidding affects loss aversion in the NFT market. Prior research offers mixed results: some suggest experience reduces loss aversion by promoting rational decision-making (e.g., [J. A. List \(2003\)](#); [Haigh and List \(2005\)](#)), while others argue experience may not diminish and could even reinforce biases (e.g., [Malmendier and Nagel \(2011\)](#); [Plott and Zeiler \(2005\)](#)). To test this, we use two proxies for experience: the number of previous placed bids or offers and the duration of market participation. Though imperfect, these measures help us assess how experience influences loss aversion.

6.1 Measure Experience by Activeness

We first consider the experience of sellers. Experience is measured by the number of previous offers a seller has made in the NFT market. For instance, if a seller has made a total of 10 offers, the first offer would be included in the first quartile of offering experience, while the 10th offer would be included in the second quartile. Specifically, we divide the sample into four groups based on the quantile distribution of offering experience, with the first, second, and third quartiles corresponding to sellers with 7, 26, and 88 previous offers, respectively. By analyzing these subsamples, we can examine whether offering experience

Table 10: Sellers' Loss Aversion by Experience(Activeness)

	[75%, 100%]	[50%, 75%]	[25%, 50%]	[0%, 25%]
<i>Panel A: Full Sample</i>				
Reference Dependence	0.04*** (0.01)	0.07*** (0.01)	0.05*** (0.01)	0.06*** (0.01)
Hedonic Price	0.95*** (0.00)	0.96*** (0.00)	0.93*** (0.00)	0.88*** (0.01)
Residual from last sale	0.13*** (0.01)	0.10*** (0.01)	0.10*** (0.02)	0.01 (0.02)
Holding period	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)
Observations	6499	6756	6711	6834
<i>Panel B: Gain Sample</i>				
Reference Dependence	0.01 (0.01)	0.04*** (0.01)	0.02 (0.01)	0.02 (0.01)
Hedonic Price	0.95*** (0.00)	0.97*** (0.00)	0.94*** (0.01)	0.90*** (0.01)
Residual from last sale	0.05*** (0.01)	0.07*** (0.01)	0.01 (0.02)	-0.04** (0.02)
Holding Period	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00** (0.00)
Observations	4752	4626	4670	4779
<i>Panel C: Loss Sample</i>				
Reference Dependence	0.18*** (0.04)	0.36*** (0.06)	0.41*** (0.07)	0.82*** (0.12)
Hedonic Price	0.94*** (0.00)	0.94*** (0.01)	0.92*** (0.01)	0.86*** (0.01)
Residual from last sale	0.78*** (0.04)	0.54*** (0.05)	0.68*** (0.05)	0.51*** (0.10)
Holding Period	0.00*** (0.00)	0.01*** (0.00)	-0.00*** (0.00)	-0.01* (0.00)
Observations	1747	2130	2041	2055

Note: The dependent variable is sellers' offering prices. All measures are in dollar. Columns (1) shows the OLS regression results using whole sample. Columns (2) – (5) shows the OLS regression results using different subsamples. Different subsamples are constructed by quartile distribution of the historical offering experience. The first to third quartile of offering experience is 7, 26 and 88. All estimates use dollar term. Standard errors in parentheses. * : ($p < 0.10$), ** : ($p < 0.05$), *** : ($p < 0.01$).

influences the degree of loss aversion exhibited by sellers.

Table 10 presents how sellers' experience in the NFT market influences loss aversion across different levels of market participation. In the full sample (Panel A), we observe a statistically significant reference dependence effect across all experience levels, ranging from 0.04 to 0.07. This suggests that sellers generally exhibit some degree of anchoring to the last transaction price, regardless of their experience level.

When comparing the gain sample (Panel B) to the loss sample (Panel C), the results reveal differences in behavior between more and less experienced sellers in the NFT market. More experienced sellers, represented by the 75-100% quartile, exhibit significantly reduced loss aversion compared to their less experienced counterparts. In the gain sample, these experienced sellers show a reference dependence coefficient of 0.01, indicating that a 10% increase in potential gains leads to only a 0.1% increase in their offering price. In the loss sample, their coefficient rises to 0.18, suggesting that a 10% increase in potential losses results in a 1.8% increase in the offering price. While this still indicates some degree of loss aversion, with a ratio of 18 ($0.18/0.01$), it is substantially lower than what we observe in less experienced sellers. For example, for less experienced sellers in the 0-25% quartile, they show a shift in their pricing behavior when facing potential losses. Their reference dependence coefficient jumps from a mere 0.02 in the gain sample to a substantial 0.82 in the loss sample. This means that while a 10% increase in potential gains only leads to a 0.2% increase in their offering price, a 10% increase in potential losses triggers an 8.2% increase in the offering price. This results in a loss aversion ratio of 41 ($0.82/0.02$), more than double that of the most experienced sellers.

The smaller gap between gain and loss coefficients for experienced sellers indicates they're less likely to overprice their NFTs when facing potential losses. For instance, the difference between their loss and gain coefficients is 0.17 ($0.18 - 0.01$), while for the least experienced sellers, this gap is a substantial 0.80 ($0.82 - 0.02$).

These findings reveal that the process of making offers in the NFT market serves as a powerful learning mechanism. With each offer made, sellers appear to be fine-tuning their pricing strategies, gradually shifting from loss-averse decisions to more rational pricing approaches. This evolution underscores the role of active market participation in developing effective selling strategies in the NFT marketplace.

We apply the same methodology to analyze the experience of bidders. Here, experience is measured by the number of previous bids placed by a bidder in the market. The cutoff points for bidding experience are defined as the 25th, 50th, and 75th percentiles,

corresponding to 3, 10, and 37 previous bids, respectively. Table 11 presents the results for bidders. The pattern of loss aversion among bidders differs from that observed in sellers. In the full sample (Panel A), we see that the most experienced bidders (those with more than 37 bids) exhibit the strongest reference dependence, with a coefficient of 0.18. This suggests that for these experienced bidders, a 10% increase in the reference price leads to a 1.8% increase in their bid price. In contrast, less experienced bidders show little to no significant reference dependence.

When comparing the gain sample (Panel B) to the loss sample (Panel C), we observe that the most experienced bidders (those with more than 37 bids) show significant reference dependence in both scenarios, but the effect is much stronger in loss situations. In the gain sample, these experienced bidders have a reference dependence coefficient of 0.15, indicating that a 10% increase in the reference price leads to a 1.5% increase in their bid price. However, in the loss sample, their coefficient jumps to 0.50, meaning a 10% increase in the reference price results in a 5% increase in their bid price. This difference (0.50 vs 0.15) suggests that experienced bidders are much more sensitive to potential losses than to potential gains. In contrast, less experienced bidders (those with fewer than 37 bids) show no significant reference dependence in either gain or loss situations. Their bidding behavior appears to be less influenced by past prices, regardless of whether they're facing potential gains or losses. This difference between experienced and less experienced bidders indicates that market experience may lead to increased loss aversion for bidders.

To address concerns about strategic bidding and offering behavior based on the actions of counterparts or competitors in an auction, we narrow our focus to the actions of the first bidder or first seller in each auction. By examining the actions of the initial initiator, who lacks information about other participants in the same auction, we can minimize the influence of strategic motives in their actions. The first action in an auction is likely to be driven by intrinsic motivations rather than strategic considerations, as the initiator has limited information about other participants. Therefore, by analyzing the behavior of the first bidder or first seller, we can reduce the potential impact of strategic decision-making on the observed results.

Table 12 presents the results of sellers' reference dependence when making the first offer in an NFT auction by their level of experience. In the Full Sample (Panel A), we observe significant reference dependence across all experience levels, ranging from 0.07 for the least experienced to 0.11 for the most experienced sellers. This suggests that even when making fresh offers, sellers tend to anchor their prices to past reference points.

Table 11: Bidders' Loss Aversion and Experience(Activeness)

	[75%, 100%]	[50%, 75%]	[25%, 50%]	[0%, 25%]
<i>Panel A: Full Sample</i>				
Reference Dependence	0.18*** (0.05)	-0.00 (0.02)	0.03 (0.03)	0.01 (0.03)
Hedonic Price	1.09*** (0.03)	1.01*** (0.01)	1.01*** (0.02)	0.97*** (0.02)
Residuals from Last Sale	-0.02 (0.08)	0.12*** (0.04)	0.24*** (0.04)	0.14*** (0.05)
Holding Period	-0.01*** (0.00)	-0.00*** (0.00)	-0.00 (0.00)	-0.00** (0.00)
Observations	1681	1665	1795	2010
<i>Panel B: Gain Sample</i>				
Reference Dependence	0.15*** (0.06)	-0.02 (0.02)	0.03 (0.02)	0.01 (0.04)
Hedonic Price	1.11*** (0.04)	1.02*** (0.02)	1.02*** (0.02)	0.98*** (0.02)
Residuals from Last Sale	-0.08 (0.10)	0.08** (0.04)	0.23*** (0.03)	0.12** (0.05)
Holding Period	-0.01*** (0.00)	-0.00*** (0.00)	-0.00 (0.00)	-0.00** (0.00)
Observations	1318	1328	1369	1517
<i>Panel C: Loss Sample</i>				
Reference Dependence	0.50*** (0.14)	0.11 (0.19)	0.21 (0.33)	-0.05 (0.36)
Hedonic Price	1.13*** (0.02)	1.00*** (0.03)	1.02*** (0.04)	0.97*** (0.05)
Residuals from Last Sale	0.29*** (0.10)	0.60*** (0.12)	0.49** (0.22)	0.69*** (0.26)
Holding Period	0.01*** (0.00)	-0.00 (0.00)	0.00 (0.00)	0.00 (0.01)
Observations	363	337	426	493

Note: The dependent variable is bidders' bidding prices. All measures are in dollar. Columns (1) shows the OLS regression results using whole sample. Columns (2) – (5) shows the OLS regression results using different subsamples. Different subsamples are constructed by quartile distribution of the historical bidding experience. The first to third quartile of bidding experience is 3, 10 and 37. All estimates use dollar term. Standard errors in parentheses. * : ($p < 0.10$), ** : ($p < 0.05$), *** : ($p < 0.01$).

Table 12: Sellers' Anchoring Effect and Experience in Fresh Actions

	All	[75%, 100%]	[50%, 75%]	[25%, 50%]	[0%, 25%]
<i>Panel A: Full Sample</i>					
Reference Dependence	0.08*** (0.01)	0.11*** (0.01)	0.10*** (0.02)	0.07*** (0.02)	0.07** (0.03)
Hedonic Price	0.91*** (0.00)	0.95*** (0.01)	0.93*** (0.01)	0.88*** (0.01)	0.84*** (0.01)
Residual from Last Sale	0.13*** (0.02)	0.29*** (0.02)	0.15*** (0.03)	0.03 (0.03)	0.07 (0.04)
Holding Period	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00 (0.00)	0.00 (0.00)
Observations	7552	1916	1971	2392	1892
<i>Panel B: Gain Sample</i>					
Reference Dependence	0.03** (0.01)	0.06*** (0.02)	0.10*** (0.02)	-0.01 (0.02)	-0.01 (0.02)
Hedonic Price	0.91*** (0.01)	0.95*** (0.01)	0.93*** (0.01)	0.90*** (0.01)	0.87*** (0.01)
Residual from Last Sale	0.05*** (0.02)	0.02 (0.03)	0.19*** (0.03)	0.05* (0.03)	-0.01 (0.03)
Holding Period	0.00** (0.00)	0.00** (0.00)	0.00*** (0.00)	0.00 (0.00)	-0.00 (0.00)
Observations	5100	1015	1032	1115	1938
<i>Panel C: Loss Sample</i>					
Reference Dependence	0.95*** (0.09)	0.56*** (0.12)	0.77*** (0.16)	0.96*** (0.13)	1.15*** (0.21)
Hedonic Price	0.92*** (0.01)	0.94*** (0.01)	0.94*** (0.01)	0.92*** (0.01)	0.86*** (0.01)
Residual from Last Sale	0.35*** (0.08)	0.38*** (0.10)	0.33** (0.15)	0.16 (0.11)	0.42** (0.19)
Holding Period	-0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)	0.00 (0.00)
Observations	2452	457	580	525	890

Note: The dependent variable is sellers' offering prices. All measures are in dollar. Columns (1) shows the OLS regression results using whole sample. Columns (2) – (5) shows the OLS regression results using different subsamples. Different subsamples are constructed by quartile distribution of the historical bidding experience. The first to third quartile of offering experience is 2, 11 and 47. All estimates use dollar term. Standard errors in parentheses. * : ($p < 0.10$), ** : ($p < 0.05$), *** : ($p < 0.01$).

Table 13: Bidders' Loss Aversion and Experience in Fresh Actions

	All	[75%, 100%]	[50%, 75%]	[25%, 50%]	[0%, 25%]
<i>Panel A: Full Sample</i>					
Reference Dependence	0.03 (0.03)	0.25*** (0.08)	0.05 (0.03)	0.03 (0.04)	0.01 (0.06)
Hedonic Price	1.00*** (0.02)	1.04*** (0.04)	1.02*** (0.02)	1.00*** (0.02)	0.96*** (0.03)
Residual from Last Sale	0.13*** (0.05)	-0.10 (0.14)	0.16*** (0.05)	0.12** (0.06)	0.11 (0.08)
Holding Period	-0.01*** (0.00)	-0.02*** (0.00)	-0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)
Observations	3499	889	942	1141	957
<i>Panel B: Gain Sample</i>					
Reference Dependence	-0.00 (0.03)	0.10 (0.06)	0.03 (0.05)	-0.02 (0.06)	0.10 (0.07)
Hedonic Price	1.01*** (0.02)	1.04*** (0.03)	1.01*** (0.02)	0.98*** (0.03)	0.99*** (0.04)
Residual from Last Sale	0.14*** (0.05)	-0.02 (0.10)	0.27*** (0.07)	0.11 (0.08)	0.04 (0.10)
Holding Period	-0.01*** (0.00)	-0.02*** (0.00)	-0.00 (0.00)	-0.00* (0.00)	-0.00 (0.00)
Observations	2649	1145	451	583	470
<i>Panel B: Gain Sample</i>					
Reference Dependence	0.26 (0.22)	0.01 (0.20)	0.53* (0.32)	1.32 (0.84)	0.15 (0.79)
Hedonic Price	1.03*** (0.03)	1.08*** (0.02)	1.10*** (0.04)	0.98*** (0.09)	0.83*** (0.13)
Residual from Last Sale	0.25 (0.17)	0.40** (0.15)	-0.04 (0.25)	0.01 (0.52)	0.41 (0.78)
Holding Period	0.00 (0.00)	0.00 (0.00)	0.01 (0.00)	0.01 (0.02)	-0.02 (0.02)
Observations	850	348	182	171	149

Note: The dependent variable is bidders' bidding prices. All measures are in dollar. Columns (1) shows the OLS regression results using whole sample. Columns (2) – (5) shows the OLS regression results using different subsamples. Different subsamples are constructed by quartile distribution of the historical bidding experience. The first to third quartile of bidding experience is 2, 8 and 35. All estimates use dollar term. Standard errors in parentheses. * : ($p < 0.10$), ** : ($p < 0.05$), *** : ($p < 0.01$).

In gain situations, the reference dependence effect is relatively weak, with coefficients ranging from -0.01 for the least experienced sellers to 0.10 for those in the 50-75% experience quartile. However, loss situations trigger a large increase in reference dependence across all experience levels, with coefficients soaring from 0.56 for the most experienced sellers to 1.15 for the least experienced. This contrast underscores that sellers, regardless of their experience, are significantly more sensitive to potential losses than to potential gains when making initial offers.

The impact of experience on this behavior is particularly noteworthy. While the reference dependence effect in loss situations decreases with experience (from 1.15 for the least experienced to 0.56 for the most experienced), it remains substantially higher than in gain situations across all experience levels. This suggests that even the most seasoned sellers exhibit strong loss aversion when making fresh offers in potential loss scenarios.

When we contrast these fresh action results with the previous findings for all seller actions, we observe both similarities and differences. In both cases, loss aversion persists, with stronger reference dependence in loss situations compared to gain situations. However, the effect is more pronounced in fresh actions, particularly for less experienced sellers. The impact of experience in mitigating reference dependence appears more dramatic in fresh actions, with a larger decrease in the coefficient from the least to most experienced sellers compared to the overall seller analysis.

We observe a clear trend of decreasing reference dependence as seller experience increases. The least experienced sellers (0-25% quartile) exhibit the strongest reference dependence with a coefficient of 1.15, indicating that a 10% increase in the reference price leads to an 11.5% increase in their offer price. This effect moderates as we move up the experience ladder. Sellers in the 25-50% experience quartile show a coefficient of 0.96, those in the 50-75% quartile have a coefficient of 0.77, and the most experienced sellers (75-100% quartile) demonstrate the lowest reference dependence with a coefficient of 0.56. This means that for the most experienced sellers, a 10% increase in the reference price results in only a 5.6% increase in their offer price - less than half the effect seen in the least experienced group.

The patterns here are consistent with the previous analysis using all auctions (not restricted to fresh actions) shown in Table 10. In both cases, we observe a decreasing trend in reference dependence as seller experience increases in loss situations. However, the effect is more pronounced in fresh actions. For instance, in Table 10, the reference dependence coefficient for the least experienced sellers in loss situations was 0.82, compared to

1.15 in fresh actions. Similarly, for the most experienced sellers, the coefficient was 0.18 in all auctions, versus 0.56 in fresh actions. This comparison suggests that while experience consistently mitigates loss aversion, the effect of experience is even more critical in fresh actions where sellers lack information about other participants. Nonetheless, in both analyses, we see that reference dependence remains significant even for the most experienced sellers, indicating that loss aversion is a persistent feature of NFT market behavior across all experience levels, though its intensity varies with experience.

Table 13 presents an analysis of bidder behavior in fresh actions by experience levels. In the Full Sample, we observe that only the most experienced bidders (75-100% quartile) exhibit significant reference dependence, with a coefficient of 0.25. This indicates that for these most experienced bidders, a 10% increase in the reference price leads to a 2.5% increase in their bid price. Interestingly, less experienced bidders show no significant anchoring to past prices in their initial bids. When we examine the Gain Sample, the reference dependence effect becomes insignificant across all experience levels, suggesting that in potential gain situations, bidders are less likely to anchor their fresh bids to past prices, regardless of their experience. The Loss Sample presents a slightly different picture, with a marginally significant reference dependence effect (0.53) for bidders in the 50-75% experience quartile, while other experience levels, including the most experienced bidders, show no significant effect.

Comparing these results to the previous analysis of all bidder actions reveals some contrasts. In the earlier analysis, more experienced bidders generally showed stronger reference dependence, particularly in loss situations. However, this pattern is less evident in fresh actions. Moreover, the magnitude of the reference dependence effect in fresh actions is generally lower than observed in the analysis of all actions. These findings suggest that when bidders make fresh bids without information about other participants, their behavior is less influenced by loss aversion compared to their overall bidding behavior.

6.2 Measure Experience by Time

We employ an alternative measure of experience based on the duration between an investor's first and current actions in the Crypto Punk market. For sellers, this is calculated as the time elapsed since their initial offering, while for bidders, it's the period since their first bid placement. This time-based metric provides a different perspective on experience, assuming that longer market participation correlates with greater expertise. Using

this measure, we categorize the entire sample into four quartiles based on the distribution of experience duration. This approach allows us to investigate how the length of market involvement might influence investor behavior and decision-making patterns.

Table 14 shows the results of seller behavior using time-based experience as a measure. In the full sample (Panel A), we observe a consistent reference dependence effect across all experience levels, with coefficients ranging from 0.07 to 0.08. This suggests that, overall, sellers tend to anchor their prices to past references regardless of how long they've been in the market. The effect is relatively stable across experience quartiles, indicating that time in the market doesn't significantly alter this general reference dependence behavior.

In the gain sample, we see a weak reference dependence effect that slightly decreases with experience, ranging from 0.03 for the most experienced sellers to 0.01 for the least experienced, with the effect not being statistically significant for the two least experienced quartiles. This suggests that in gain situations, the length of time in the market has minimal impact on sellers' tendency to anchor their prices to past references.

The loss sample, however, reveals a much stronger reference dependence effect that increases as experience decreases. The coefficient for the most experienced sellers is 0.36, rising sharply to 1.01 for the least experienced sellers. This indicates that when facing potential losses, sellers with less time in the market exhibit much stronger anchoring behavior, with a 10% increase in the reference price leading to a 10.1% increase in their offer price for the least experienced group.

This pattern is consistent with the results we observed in Table 10, where experience was measured by activeness (number of offers). In both analyses, we see that more experienced sellers exhibit less reference dependence in loss situations. These consistent results across different measures of experience provide robust insights into the nature of learning and behavior in the NFT market. Both active participation and longer market presence seem to help sellers develop strategies that mitigate loss aversion.

Table 15 presents the results of bidder loss aversion using time-based experience as a measure. Examining the full sample, we observe a varying pattern of reference dependence across experience levels. The most experienced bidders show a significant positive reference dependence (0.08), while the second most experienced group shows a weaker effect (0.03). Interestingly, the second least experienced group exhibits a significant negative reference dependence (-0.16), while the least experienced group shows no significant effect. This suggests that the relationship between time-based experience and reference dependence is not linear for bidders. When comparing the gain sample to the loss sample,

we notice some differences. In the gain sample, the pattern is similar to the full sample, with the most experienced bidders showing positive reference dependence (0.07) and the second least experienced group showing negative reference dependence (-0.22). However, in the loss sample, we observe no statistically significant reference dependence across all experience levels. This suggests that when facing potential losses, bidders across all experience levels are less likely to be influenced by past prices and may be more focused on current market valuations. These findings contrast with our previous results for sellers and with the results we observed when measuring bidder experience by activity level. The length of time in the market might shape bidder behavior in ways that differ from mere repetition of bidding activity.

7 Conclusion

The Non-Fungible Token (NFT) market offers a unique opportunity for economists to study behavioral biases exhibited by investors. This paper focused on investigating two specific biases: the anchoring effect and loss aversion. We aimed to examine the presence and impact of these biases within this emerging financial market. Our findings confirm the existence of both anchoring effects and loss aversion in the NFT market. Through the analysis of bidding and offering data, we were able to identify these behavioral biases on both the bidder and seller sides of the market. The results clearly demonstrate that both bidders and sellers are susceptible to loss aversion and anchoring effects. Moreover, our analysis revealed that the severity of these biases is influenced by the reference point (anchor) used. The more recent the reference point, the more pronounced the loss aversion and anchoring effects become. Additionally, when we focused on agents facing potential losses, such as sellers, the magnitude of these biases was even more significant. We interpret this as evidence of sellers' loss aversion and the endogenous generation of loss aversion among bidders.

In addition, we explored the role of experience as a potential moderator of loss aversion and anchoring effects in the NFT market. Interestingly, our findings suggest that experience does not serve as a qualified moderator for these biases. Experienced bidders tend to exhibit even stronger reference dependence compared to inexperienced bidders, indicating that experience may reinforce behavioral biases rather than mitigate them. On the other hand, the behavior of experienced sellers is similar to that of inexperienced sellers, suggesting that experience does not significantly impact their susceptibility to biases. Fi-

Table 14: Sellers' Loss Aversion and Experience (Time)

	[75%, 100%]	[50%, 75%]	[25%, 50%]	[0%, 25%]
<i>Panel A: Full Sample</i>				
Reference Dependence	0.07*** (0.01)	0.07*** (0.01)	0.08*** (0.02)	0.07*** (0.01)
Hedonic Price	0.97*** (0.00)	0.96*** (0.00)	0.93*** (0.00)	0.88*** (0.01)
Residual from Last Sale	0.17*** (0.01)	0.09*** (0.01)	-0.03 (0.02)	0.07*** (0.02)
Holding Period	0.00*** (0.00)	0.00*** (0.00)	0.01*** (0.00)	0.00*** (0.00)
Observations	6501	6476	6496	6463
<i>Panel B: Gain Sample</i>				
Reference Dependence	0.03*** (0.01)	0.06*** (0.01)	0.03 (0.02)	0.01 (0.01)
Hedonic Price	0.97*** (0.00)	0.97*** (0.00)	0.94*** (0.01)	0.89*** (0.01)
Residual from Last Sale	0.06*** (0.01)	0.05*** (0.01)	-0.05** (0.02)	-0.03 (0.02)
Holding Period	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00** (0.00)
Observations	5410	4459	4450	3922
<i>Panel C: Loss Sample</i>				
Reference Dependence	0.36*** (0.08)	0.15*** (0.04)	0.66*** (0.07)	1.01*** (0.12)
Hedonic Price	0.95*** (0.01)	0.93*** (0.00)	0.94*** (0.01)	0.88*** (0.01)
Residual from Last Sale	0.80*** (0.05)	0.76*** (0.04)	0.24*** (0.06)	0.33*** (0.11)
Holding Period	0.00 (0.00)	-0.00 (0.00)	-0.00* (0.00)	-0.00 (0.00)
Observations	1091	2017	2046	2541

Note: The dependent variable is sellers' offering prices. All measures are in dollar. Columns (1) shows the OLS regression results using whole sample. Columns (2) – (5) shows the OLS regression results using different subsamples. Different subsamples are constructed by quartile distribution of the historical offering experience. The first to third quartile of offering experience is 7, 26 and 88. All estimates use dollar term. Standard errors in parentheses. * : ($p < 0.10$), ** : ($p < 0.05$), *** : ($p < 0.01$).

Table 15: Bidders' Loss Aversion and Experience (Time)

	[75%, 100%]	[50%, 75%]	[25%, 50%]	[0%, 25%]
<i>Panel A: Full Sample</i>				
Reference Dependence	0.08*** (0.02)	0.03* (0.01)	-0.16*** (0.05)	0.04 (0.04)
Hedonic Price	1.01*** (0.01)	1.02*** (0.01)	1.10*** (0.03)	0.97*** (0.02)
Residual from Last Sale	0.05 (0.03)	0.25*** (0.02)	0.27*** (0.08)	0.10 (0.06)
Holding Period	0.00 (0.00)	-0.00** (0.00)	-0.02*** (0.00)	-0.00* (0.00)
Observations	1644	1648	1647	1637
<i>Panel B: Gain Sample</i>				
Reference Dependence	0.07*** (0.02)	0.01 (0.02)	-0.22*** (0.06)	0.04 (0.05)
Hedonic Price	1.02*** (0.01)	1.01*** (0.01)	1.15*** (0.04)	0.98*** (0.03)
Residual from Last Sale	-0.00 (0.03)	0.24*** (0.02)	0.22** (0.09)	0.09 (0.06)
Holding Period	0.00 (0.00)	-0.00*** (0.00)	-0.02*** (0.00)	-0.00* (0.00)
Observations	1267	1293	1293	1248
<i>Panel C: Loss Sample</i>				
Reference Dependence	0.37 (0.23)	0.16 (0.11)	0.04 (0.27)	-0.11 (0.43)
Hedonic Price	1.02*** (0.04)	1.08*** (0.02)	1.05*** (0.03)	1.00*** (0.06)
Residual from Last Sale	0.35** (0.16)	0.48*** (0.09)	0.66*** (0.17)	0.54 (0.33)
Holding Period	0.00 (0.00)	0.01*** (0.00)	0.00 (0.00)	0.00 (0.01)
Observations	377	355	354	389

Note: the dependent variable is bidders' bidding prices. All measures are in dollar. Columns (1) shows the OLS regression results using whole sample. Columns (2) – (5) shows the OLS regression results using different subsamples. Different subsamples are constructed by quartile distribution of experience. Standard errors in parentheses. * : ($p < 0.10$), ** : ($p < 0.05$), *** : ($p < 0.01$).

nally, we examined whether sellers and bidders strategically respond to the actions of their counterparts in the market. Our analysis did not find evidence to support this notion, indicating that both sellers and bidders are indeed influenced by the anchoring effect and loss aversion.

References

- Bao, H., & Roubaud, D. (2021, December). Recent Development in Fintech: Non-Fungible Token. *FinTech*, 1(1), 44–46. Retrieved 2022-03-18, from <https://www.mdpi.com/2674-1032/1/1/3> doi: 10.3390/fintech1010003
- Beggs, A., & Graddy, K. (1997). Declining Values and the Afternoon Effect: Evidence from Art Auctions. *The RAND Journal of Economics*, 28(3), 544. Retrieved 2022-03-23, from <http://doi.wiley.com/10.2307/2556028> doi: 10.2307/2556028
- Beggs, A., & Graddy, K. (2009, May). Anchoring Effects: Evidence from Art Auctions. *American Economic Review*, 99(3), 1027–1039. Retrieved 2022-03-18, from <https://pubs.aeaweb.org/doi/10.1257/aer.99.3.1027> doi: 10.1257/aer.99.3.1027
- Bokhari, S., & Geltner, D. (2011, December). Loss Aversion and Anchoring in Commercial Real Estate Pricing: Empirical Evidence and Price Index Implications: Loss Aversion and Anchoring in Commercial Real Estate Pricing. *Real Estate Economics*, 39(4), 635–670. Retrieved 2022-03-18, from <https://onlinelibrary.wiley.com/doi/10.1111/j.1540-6229.2011.00308.x> doi: 10.1111/j.1540-6229.2011.00308.x
- Borri, N., Liu, Y., & Tsyvinski, A. (2022). The Economics of Non-Fungible Tokens. *SSRN Electronic Journal*. Retrieved 2022-06-05, from <https://www.ssrn.com/abstract=4052045> doi: 10.2139/ssrn.4052045
- Bracke, P., & Tenreyro, S. (2021, April). History Dependence in the Housing Market. *American Economic Journal: Macroeconomics*, 13(2), 420–443. Retrieved 2022-03-18, from <https://pubs.aeaweb.org/doi/10.1257/mac.20180241> doi: 10.1257/mac.20180241
- Brown, A. L., Imai, T., Vieider, F. M., & Camerer, C. F. (2024). Meta-analysis of empirical estimates of loss aversion. *Journal of Economic Literature*, 62(2), 485–516.
- Chapman, J., Snowberg, E., Wang, S., & Camerer, C. (2024). Looming large or seeming small? attitudes towards losses in a representative sample. *Review of Economic Studies*, rdae093.

- Dimson, E., Rousseau, P. L., & Spaenjers, C. (2015, November). The price of wine. *Journal of Financial Economics*, 118(2), 431–449. Retrieved 2022-03-18, from <https://linkinghub.elsevier.com/retrieve/pii/S0304405X15001440> doi: 10.1016/j.jfineco.2015.08.005
- Dimson, E., & Spaenjers, C. (2011, May). Ex post: The investment performance of collectible stamps. *Journal of Financial Economics*, 100(2), 443–458. Retrieved 2022-03-18, from <https://linkinghub.elsevier.com/retrieve/pii/S0304405X10002941> doi: 10.1016/j.jfineco.2010.12.005
- Dougal, C., Engelberg, J., & Parsons, C. A. (2015). Anchoring on Credit Spreads. *The Journal of Finance*, 70(3), 1039–1080.
- Fixler, D., Fortuna, C., Greenlees, J., & Lane, W. (1999). The use of hedonic regressions to handle quality change: The experience in the us cpi. In *fifth meeting of the international working group on price indices, reykjavik, iceland, august*. bureau of labor statistics. <http://www.statice.is/ottawa/bls.pdf> (pp. 1–20).
- Gal, D., & Rucker, D. D. (2018). The loss of loss aversion: Will it loom larger than its gain? *Journal of Consumer Psychology*, 28(3), 497–516.
- Genesove, D., & Mayer, C. (2001). LOSS AVERSION AND SELLER BEHAVIOR: EVIDENCE FROM THE HOUSING MARKET. *Quarterly Journal of Economics*, 116(4), 1233–1260.
- Graddy, K., Loewenstein, L., Mei, J., Moses, M., & Pownall, R. (2015). Empirical Evidence of Anchoring and Loss Aversion from Art Auctions. *SSRN Electronic Journal*.
- Guo, S. (2024). Reference price updating in the housing market. *Working Papers*.
- Gurdgiev, C., & O’Loughlin, D. (2020, March). Herding and anchoring in cryptocurrency markets: Investor reaction to fear and uncertainty. *Journal of Behavioral and Experimental Finance*, 25, 100271. Retrieved 2022-03-18, from <https://linkinghub.elsevier.com/retrieve/pii/S2214635019301534> doi: 10.1016/j.jbef.2020.100271
- Haigh, M. S., & List, J. A. (2005, February). Do Professional Traders Exhibit Myopic Loss Aversion? An Experimental Analysis. *The Journal of Finance*, 60(1), 523–534. Retrieved 2022-03-23, from <https://onlinelibrary.wiley.com/doi/10.1111/j.1540-6261.2005.00737.x> doi: 10.1111/j.1540-6261.2005.00737.x
- Horky, F., Rachel, C., & Fidrmuc, J. (2022). Price Dynamics of Non-Fungible Tokens: The Case of the Digital Arts Market. *SSRN Electronic Journal*. Retrieved 2022-06-05, from <https://www.ssrn.com/abstract=4080372> doi: 10.2139/ssrn.4080372

- Ivanova-Stenzel, R., & Seres, G. (2021, June). Are strategies anchored? *European Economic Review*, 135, 103725. Retrieved 2022-03-18, from <https://linkinghub.elsevier.com/retrieve/pii/S0014292121000787> doi: 10.1016/j.eurocorev.2021.103725
- John A. List. (2004). Neoclassical Theory versus Prospect Theory: Evidence from the Marketplace. *Econometrica*, 72(2), 615–625.
- Jullien, B., & Salanie, B. (2000). Estimating Preferences under Risk: The Case of Racetrack Bettors. *Journal of Political Economy*, 108(3), 503–530.
- Kahneman, D., Knetsch, J. L., & Thaler, R. H. (1990). Experimental Tests of the Endowment Effect and the Coase Theorem. *Journal of Political Economy*, 98(6), 1325–1348.
- Kahneman, D., & Tversky, A. (1974). Judgment under uncertainty: Heuristics and biases: Biases in judgments reveal some heuristics of thinking under uncertainty. *science*, 185(4157), 1124–1131.
- Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 363–391.
- Kong, D.-R., & Lin, T.-C. (2021). Alternative investments in the Fintech era: The risk and return of Non-Fungible Token (NFT). *SSRN Electronic Journal*. Retrieved 2022-03-18, from <https://www.ssrn.com/abstract=3914085> doi: 10.2139/ssrn.3914085
- Kőszegi, B., & Rabin, M. (2006). A MODEL OF REFERENCE-DEPENDENT PREFERENCES. *QUARTERLY JOURNAL OF ECONOMICS*, CXXI(4), 33.
- Lange, A., & Ratan, A. (2010, March). Multi-dimensional reference-dependent preferences in sealed-bid auctions – How (most) laboratory experiments differ from the field. *Games and Economic Behavior*, 68(2), 634–645. Retrieved 2022-03-18, from <https://linkinghub.elsevier.com/retrieve/pii/S0899825609001699> doi: 10.1016/j.geb.2009.08.008
- List, J., & Price, M. (2013, August). *Using Field Experiments in Environmental and Resource Economics* (Tech. Rep. No. w19289). Cambridge, MA: National Bureau of Economic Research. Retrieved 2022-03-23, from <http://www.nber.org/papers/w19289.pdf> doi: 10.3386/w19289
- List, J. A. (2003, February). Does Market Experience Eliminate Market Anomalies? *The Quarterly Journal of Economics*, 118(1), 41–71. Retrieved 2022-03-23, from <https://academic.oup.com/qje/article-lookup/doi/10.1162/00335530360535144> doi: 10.1162/00335530360535144
- Lovo, S., & Spaenjers, C. (2018, March). A Model of Trading in the Art Market. *Amer-*

- ican Economic Review*, 108(3), 744–774. Retrieved 2022-03-18, from <https://pubs.aeaweb.org/doi/10.1257/aer.20160522> doi: 10.1257/aer.20160522
- Malmendier, U., & Nagel, S. (2011). Depression babies: do macroeconomic experiences affect risk taking? *The quarterly journal of economics*, 126(1), 373–416.
- Mandel, B. R. (2009, August). Art as an Investment and Conspicuous Consumption Good. *American Economic Review*, 99(4), 1653–1663. Retrieved 2022-03-18, from <https://pubs.aeaweb.org/doi/10.1257/aer.99.4.1653> doi: 10.1257/aer.99.4.1653
- Mei, J., & Moses, M. (2002). Art as an Investment and the Underperformance of Masterpieces. *American Economic Review*, 92(5), 1656–1668.
- Mrkva, K., Johnson, E. J., Gächter, S., & Herrmann, A. (2020, July). Moderating Loss Aversion: Loss Aversion Has Moderators, But Reports of its Death are Greatly Exaggerated. *Journal of Consumer Psychology*, 30(3), 407–428. Retrieved 2022-03-23, from <https://onlinelibrary.wiley.com/doi/10.1002/jcpy.1156> doi: 10.1002/jcpy.1156
- Nadini, M., Alessandretti, L., Di Giacinto, F., Martino, M., Aiello, L. M., & Baronchelli, A. (2021, December). Mapping the NFT revolution: market trends, trade networks, and visual features. *Scientific Reports*, 11(1), 20902. Retrieved 2022-03-18, from <https://www.nature.com/articles/s41598-021-00053-8> doi: 10.1038/s41598-021-00053-8
- Northcraft, G. B., & Neale, M. A. (1987, February). Experts, amateurs, and real estate: An anchoring-and-adjustment perspective on property pricing decisions. *Organizational Behavior and Human Decision Processes*, 39(1), 84–97. Retrieved 2022-03-18, from <https://linkinghub.elsevier.com/retrieve/pii/074959788790046X> doi: 10.1016/0749-5978(87)90046-X
- Oh, S., Rosen, S., & Zhang, A. L. (2022). Investor Experience Matters: Evidence from Generative Art Collections on the Blockchain. *SSRN Electronic Journal*. Retrieved 2022-06-05, from <https://www.ssrn.com/abstract=4042901> doi: 10.2139/ssrn.4042901
- Orley Ashenfelter, & Kathryn Graddy. (2003). Auctions and the Price of Art. *Journal of Economic Literature*, 41(3), 763–787.
- Plott, C. R., & Zeiler, K. (2005). The willingness to pay–willingness to accept gap, the “endowment effect,” subject misconceptions, and experimental procedures for eliciting valuations. *American Economic Review*, 95(3), 530–545.
- Pope, D. G., & Schweitzer, M. E. (2011). Is Tiger Woods Loss Averse? Persistent Bias in the

- Face of Experience, Competition, and High Stakes. *American Economic Review*, 101(1), 129–157.
- Rutherford, R., Springer, T., & Yavas, A. (2005, June). Conflicts between principals and agents: evidence from residential brokerage. *Journal of Financial Economics*, 76(3), 627–665. Retrieved 2022-03-24, from <https://linkinghub.elsevier.com/retrieve/pii/S0304405X05000218> doi: 10.1016/j.jfineco.2004.06.006
- Shunda, N. (2009, November). Auctions with a buy price: The case of reference-dependent preferences. *Games and Economic Behavior*, 67(2), 645–664. Retrieved 2022-03-18, from <https://linkinghub.elsevier.com/retrieve/pii/S0899825609000761> doi: 10.1016/j.geb.2009.04.013
- Sopranzetti, B. J. (2010). Hedonic regression analysis in real estate markets: a primer. *Handbook of quantitative finance and risk management*, 1201–1207.
- Thakral, N., & Tô, L. T. (2021). Daily labor supply and adaptive reference points. *American Economic Review*, 111(8), 2417–2443.
- Wang, Q., Li, R., Wang, Q., & Chen, S. (2021, October). Non-Fungible Token (NFT): Overview, Evaluation, Opportunities and Challenges. *arXiv:2105.07447 [cs]*. Retrieved 2022-03-18, from <http://arxiv.org/abs/2105.07447> (arXiv: 2105.07447)
- Wyburn, J., & Roach, P. A. (2012). An hedonic analysis of american collectable comic-book prices. *Journal of Cultural Economics*, 36, 309–326.
- Yasseri, T., & Reher, J. (2022, January). Fooled by facts: quantifying anchoring bias through a large-scale experiment. *Journal of Computational Social Science*. Retrieved 2022-03-18, from <https://link.springer.com/10.1007/s42001-021-00158-0> doi: 10.1007/s42001-021-00158-0
- Yechiam, E. (2019). Acceptable losses: The debatable origins of loss aversion. *Psychological research*, 83(7), 1327–1339.
- Zeif, D., & Yechiam, E. (2022). Loss aversion (simply) does not materialize for smaller losses. *Judgment and Decision Making*, 17(5), 1015–1042.

Online Appendix

A Lasso Regression

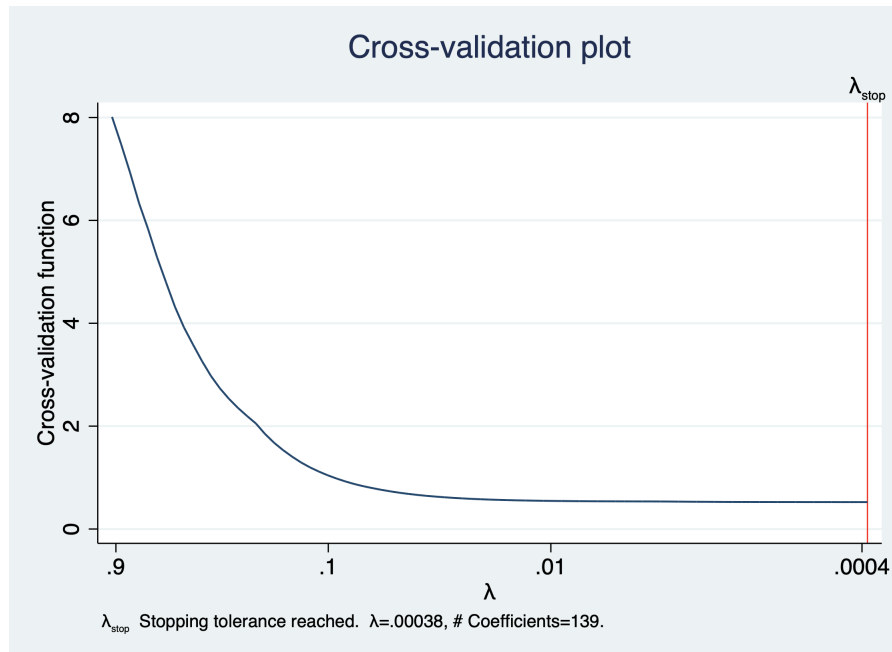


Figure 3: Lasso Regression

Table 16: Lasso vs. OLS

Name	MSE	R-squared	Obs
ols	0.51	0.94	14600
cv	0.51	0.94	14600

Note: This table shows the outcome comparison between OLS regression and Lasso regression. The result shows that there is no significant difference between OLS and Lasso.

B Using Ethereum Price to Test Anchoring Effects

For our robustness check, we also examined how prices in the NFT market, specifically sale prices, offering prices, and bidding prices, anchor on previous sale prices when measured in terms of Ethereum. The results, as shown in Table 8, Table 9, and Table 10, are qualitatively and quantitatively similar to the analyses discussed earlier. We continue to observe that prices adjust towards the previous sale price, and the impact of the previous sale price becomes more pronounced as the holding period becomes shorter.

Additionally, the more prominent effects observed in the loss sample suggest that previous sale prices have a stronger influence when the item is expected to result in a loss. Notably, bidders' self-generated loss aversion becomes more noticeable when prices are measured in terms of Ethereum. These findings provide further support for the anchoring effects and loss aversion in the NFT market, reinforcing the robustness of our earlier analyses. The use of Ethereum as the measurement unit reinforces the consistent patterns observed in pricing behavior in the NFT market.

Table 17: Price Anchoring Effect in Ethereum

	(1) All	(2) < 22 weeks	(3) < 6 weeks	(4) < 2 weeks
<i>Panel A: Full Sample</i>				
Reference Dependence	0.10*** (0.01)	0.20*** (0.01)	0.34*** (0.02)	0.72*** (0.05)
Hedonic Price	1.00*** (0.00)	0.99*** (0.00)	0.97*** (0.00)	0.96*** (0.01)
Residuals from Last Sale	0.21*** (0.01)	0.18*** (0.02)	0.15*** (0.03)	-0.03 (0.05)
Holding Period	0.00*** (0.00)	-0.00 (0.00)	0.02*** (0.00)	0.04*** (0.01)
Observations	8743	6641	4559	2971
<i>Panel B: Gain Sample</i>				
Reference Dependence	0.06*** (0.01)	0.14*** (0.02)	0.30*** (0.03)	0.59*** (0.05)
Hedonic Price	1.00*** (0.00)	1.00*** (0.00)	0.97*** (0.00)	0.97*** (0.01)
Residuals from Last Sale	0.07*** (0.01)	0.05** (0.02)	-0.03 (0.03)	-0.06 (0.05)
Holding Period	0.00*** (0.00)	0.00*** (0.00)	0.05*** (0.00)	0.08*** (0.01)
Observations	6761	4752	3136	1967
<i>Panel C: Loss Sample</i>				
Reference Dependence	0.36*** (0.04)	0.64*** (0.05)	0.85*** (0.07)	1.04*** (0.12)
Hedonic Price	0.98*** (0.01)	0.99*** (0.01)	0.99*** (0.01)	0.97*** (0.01)
Residuals from Last Sale	0.45*** (0.03)	0.21*** (0.04)	0.07 (0.06)	-0.12 (0.11)
Holding Period	-0.00*** (0.00)	-0.01*** (0.00)	-0.05*** (0.01)	-0.03 (0.03)
Observations	1982	1889	1423	1004

Note: The dependent variable is the items' sale prices. All measures are in Ethereum. Panel A shows the OLS regression results using sale prices. Panel B shows the OLS regression results using sale prices when predicted prices are estimated lower than previous sale price. Panel C shows the OLS regression results using sale prices when predicted prices are estimated lower than previous sale price. Different samples are constructed by quartile distribution of the holding period. The first to third quartile of holding period is 1, 5 and 21 weeks. All estimates use dollar term. Standard errors in parentheses. * : ($p < 0.10$), ** : ($p < 0.05$), *** : ($p < 0.01$).

Table 18: Sellers' Anchoring Effect in Ethereum

	(1) All	(2) < 16 weeks	(3) < 4 weeks	(4) < 2 weeks
<i>Panel A: Full Sample</i>				
Reference Dependence	0.10*** (0.01)	0.04** (0.01)	0.36*** (0.03)	0.62*** (0.05)
Hedonic Price	0.90*** (0.00)	0.89*** (0.00)	0.86*** (0.00)	0.84*** (0.00)
Residuals from Last Sale	0.31*** (0.01)	0.34*** (0.02)	0.08*** (0.03)	-0.05 (0.05)
Holding Period	0.00*** (0.00)	-0.01*** (0.00)	-0.02*** (0.01)	-0.04*** (0.01)
Observations	25876	19633	13677	9962
<i>Panel B: Gain Sample</i>				
Reference Dependence	0.05*** (0.01)	-0.07*** (0.02)	0.23*** (0.03)	0.31*** (0.06)
Hedonic Price	0.91*** (0.00)	0.90*** (0.00)	0.88*** (0.00)	0.87*** (0.01)
Residuals from Last Sale	0.08*** (0.01)	0.10*** (0.02)	-0.21*** (0.03)	-0.21*** (0.06)
Holding Period	0.00*** (0.00)	0.00 (0.00)	0.04*** (0.01)	0.03** (0.02)
Observations	17738	12043	8285	5871
<i>Panel C: Loss Sample</i>				
Reference Dependence	0.43*** (0.04)	0.63*** (0.04)	0.92*** (0.07)	1.17*** (0.11)
Hedonic Price	0.88*** (0.00)	0.89*** (0.00)	0.88*** (0.01)	0.84*** (0.01)
Residuals from Last Sale	0.63*** (0.03)	0.43*** (0.04)	0.23*** (0.06)	0.01 (0.10)
Holding Period	0.00 (0.00)	-0.03*** (0.00)	-0.12*** (0.01)	-0.14*** (0.02)
Observations	8138	7590	5392	4091

Note: The dependent variable is sellers' offering prices. All measures are in Ethereum. Panel A shows the OLS regression results using sale prices. Panel B shows the OLS regression results using sale prices when predicted prices are estimated lower than previous sale price. Panel C shows the OLS regression results using sale prices when predicted prices are estimated lower than previous sale price. Different samples are constructed by quartile distribution of the holding period. The first to third quartile of holding period is 1, 3 and 15 weeks. All estimates use dollar term. Standard errors in parentheses. *: ($p < 0.10$), **: ($p < 0.05$), ***: ($p < 0.01$).

Table 19: Bidders' Anchoring Effect in Ethereum

	(1) All	(2) < 47 weeks	(3) < 10 weeks	(4) < 3 weeks
<i>Panel A: Full Sample</i>				
Reference Dependence	0.09*** (0.01)	0.13*** (0.03)	0.24*** (0.05)	0.36*** (0.11)
Hedonic Price	1.00*** (0.01)	1.01*** (0.01)	1.02*** (0.01)	1.04*** (0.01)
Residuals from Last Sale	0.26*** (0.02)	0.24*** (0.03)	0.13** (0.06)	0.11 (0.11)
Holding Period	0.00 (0.00)	0.00 (0.00)	0.01 (0.01)	0.01 (0.02)
Observations	6418	4875	3325	1836
<i>Panel B: Gain Sample</i>				
Reference Dependence	0.07*** (0.02)	0.15*** (0.03)	0.22*** (0.06)	0.44*** (0.10)
Hedonic Price	1.00*** (0.01)	1.00*** (0.01)	1.00*** (0.01)	0.98*** (0.02)
Residuals from Last Sale	0.22*** (0.02)	0.20*** (0.04)	0.17*** (0.06)	0.12 (0.10)
Holding Period	-0.00 (0.00)	0.00** (0.00)	0.02*** (0.01)	0.03 (0.02)
Observations	4912	3442	2268	1503
<i>Panel C: Loss Sample</i>				
Reference Dependence	0.35*** (0.11)	0.42*** (0.12)	0.63*** (0.18)	0.66** (0.28)
Hedonic Price	1.05*** (0.02)	1.05*** (0.02)	1.08*** (0.02)	1.06*** (0.03)
Residuals from Last Sale	0.31*** (0.08)	0.26*** (0.08)	-0.07 (0.14)	-0.24 (0.23)
Holding Period	0.00 (0.00)	-0.00 (0.00)	-0.04*** (0.01)	-0.02 (0.02)
Observations	1506	1454	1166	705

Note: The dependent variable is bidders' bidding prices. All measures are in Ethereum. Panel A shows the OLS regression results using sale prices. Panel B shows the OLS regression results using sale prices when predicted prices are estimated lower than previous sale price. Panel C shows the OLS regression results using sale prices when predicted prices are estimated lower than previous sale price. Different samples are constructed by quartile distribution of the holding period. The first to third quartile of holding period is 2, 9 and 46 weeks. All estimates use dollar term. Standard errors in parentheses. * : ($p < 0.10$), ** : ($p < 0.05$), *** : ($p < 0.01$).

C Placebo Test

To further validate the effects identified in this paper, we conducted a series of placebo tests. In these tests, we randomly assigned a made-up sale price, denoted as P_r , to each item in the dataset. This means that in Equation (2), we replaced P_{is} with a randomly selected price P_r from the list of all sales prices. All other components of Equation (2) remained the same. Subsequently, we examined how the difference between P_r and the item's hedonic prediction influenced the sale, offering, and bidding prices. We replicated these tests independently 1000 times, and the distributions of the resulting coefficients for each test are plotted in Figure 1.

As depicted in the figure, the coefficients from the placebo tests for sale, offering, and bidding prices are centered around zero. This observation suggests that the anchoring effects identified in this paper are not merely a result of chance. The consistent distribution of coefficients across the placebo tests provides further evidence of the validity and robustness of the anchoring effects observed in the analysis.